



3D-Based Design Environments and Common Data Environments for Sustainable Heritage Interventions: A Case Study in German Castle Ruins

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Abstract

The increasing availability of high-resolution 3D documentation has significantly improved the recording of cultural heritage sites. However, establishing structured documentation workflows that ensure long-term data management, interoperability, and effective reuse remains a major challenge. This paper presents a documentation-centred workflow demonstrating how a Common Data Environment (CDE) can serve as the core infrastructure for managing heritage information throughout successive project phases, from documentation to design. The research focuses on ruined medieval castles in Germany, which are located in forested areas and are particularly difficult to preserve due to their exposed condition, continuous environmental degradation, and limited financial and human resources available for their maintenance. In this context, the proposed methodology explores how digital documentation and structured information management can contribute to preserving both cultural heritage and the surrounding natural environment through informed, minimally invasive interventions.

The study investigates the digital documentation of a historic castle ruin through a two-phase HBIM workflow. Rather than focusing solely on geometric modelling, the research addresses how structured information management, standardised documentation practices, and collaborative data environments support sustainable heritage management and future interventions.

The first phase consisted of comprehensive heritage documentation. The ruin was surveyed using terrestrial laser scanning and photogrammetry, producing a high-resolution point cloud dataset in E57 format. The data were processed in PointCab and interpreted to develop an HBIM model in Archicad. Throughout the modelling process, close collaboration with conservation specialists ensured that material characteristics, construction phases, deterioration, and conservation-related

observations were incorporated into the information model. Particular emphasis was placed on documentation management through the implementation of a structured Common Data Environment using Catenda. The CDE incorporated predefined folder structures, information containers, naming conventions, revision control procedures, document status management, and an identification system for digital heritage assets, establishing a consistent project information infrastructure aligned with BIM information management principles.

The second phase focused on the controlled reuse of the documented heritage information within an architectural design studio. The HBIM model was delivered to student design teams together with project documentation developed according to BIM management principles, including Exchange Information Requirements, a BIM Execution Plan, and predefined information delivery procedures. Design activities followed conservation-oriented principles based on Reduce/Reuse/Recycle (RRR) strategies, recommendations for simple and reversible construction methods, and guidelines published by the Bund Deutscher Architektinnen und Architekten for resource-conscious architectural interventions. The CDE functioned as the collaborative platform supporting information exchange, version control, review procedures, and project coordination throughout the design process.

The study demonstrates that structured server-based documentation environments extend the value of heritage recording beyond archival purposes by enabling reliable information reuse across project stages. The combination of HBIM, standardised information management, and Common Data Environments improved documentation consistency, facilitated collaboration between surveyors, conservation professionals, and designers, and provided transparent project governance throughout the lifecycle of the intervention. The findings highlight the importance of documentation standards, information management protocols, and interoperable digital workflows as essential components of sustainable cultural heritage management.

The proposed two-phase methodology positions the Common Data Environment as a long-term documentation infrastructure rather than merely a file repository. Future work will expand the workflow through collaboration with volunteers from Jugendbauhütten Rheinland-Pfalz, supporting the development of the project information model through degradation mapping, documentation of conservation measures, and integration of site records into the HBIM/CDE environment in coordination with heritage authorities and experts. This ongoing development will further support lifecycle documentation, evidence-based decision-making, and future semantic enrichment through mapping to heritage ontologies and metadata standards, including CIDOC CRM, ISO 19650-based information management frameworks, and FAIR data principles.

References

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